Feedback Systems

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Abstract

A note on the book Feedback Systems

1 Introduction

Basic feedback loop: Sensing, computing and acting

In genereal, the feedback system will have some inputs, states and outputs. Illustrated by

1. Input : $u(t)$

2. states : $x(t)$
3. outputs: $y(t)$

2 Nyquist stability criterion

The Nyquist stability criterion is a method for determining the stability of a closed loop control system. Or a feedback system that compares the output to a reference.

3 Feedback loop and stability

Before we get to the Nyquist stability criterion, we need to establish the motivation behind exploring stability. Good performance needs high-gain controllers. Yet, the feedback loop will become unstable if the gain is too high. This is where the Nyquist criterion comes in, it allows us to analyze stability.

Nyquist criterion involves

- open-loop frequency response function
- magnitude
- phase

Two approaches

1. Classical approach in the frequency domain : Single input, single output (SISO)
2. state-space models : multiple inputs, multiple outputs (MIMO)

3.1 Standard feedback loop

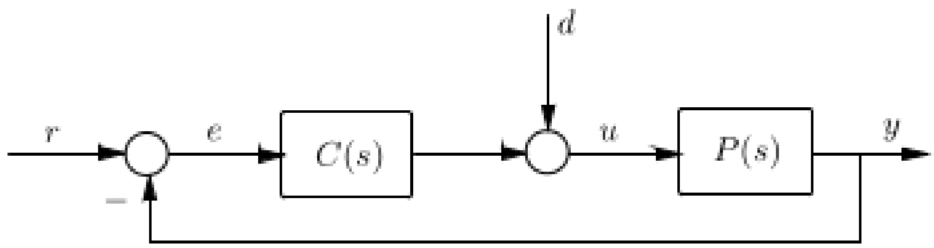

Figure 4.8: Standard feedback loop.

- r: the reference input, we want the output y to converge towards r
- e: the difference between reference and input.
- C(s): The controller transfer function. Takes input error

- d: A small unpredictable disturbance to the system, like a gust of wind for a drone
- u: the plant input.
- P(s): The plant transfer function

3.1.1 closed loop transfer functions

3.2 Principle of the argument

we have a $z \in \mathbb{C}$

$$z = a + iw$$

Then the argument is the angle of that complex number

3.3 Nyquist stability criterion

Nyquist stability criterion determines the stability of a closed loop system based on the Nyquist plot. It states that the stability depends on the number of encirclements of the critical point (-1,0) in the Nyquist plot of the open loop transfer function.

A closed loop system is stable if the number of encirclements of $-1 + 0i = N$ in the Nyquist plot is

$$N = P - Z$$

where

- $\boldsymbol{P}$: Open loop poles in the right half plane (RHP)
- $\boldsymbol{Z}$: Number of closed-loop poles in the RHP

Difference between open-loop poles & closed-loop poles:

Definition 3.1 (closed-loop pole)**.** *The position of the poles (or eigenvalues) of a closed-loop transfer function in the s-plane*

Definition 3.2 (open-loop pole)**.** *The rootsof the open-loop transfer functions denominator*

3.4 Applications of Nyquist plot

1. Analysis of closed loop stability
2. Evaluating ***stability margins*** and ***phase margins***
3. Designing PID controllers and compensators in control systems

3.4.1 The Nyquist criterion

Let n denite the number of poles of $P(s)C(s)$ in $0 \leq Re(s)$. Construct the Nyquist plot of $P(s)C(s)$, indenting to the right around poles on the imag. axis. Then the feedback system is stable iff the Nyquist plot ***doesnt pass through*** $\frac{-1}{K}$ ***and encircles it exactly n times counterclockwise***

So what range of K is the system stable then?

4 Optimal Control

In general we have a time invariant continuous system

$$\dot{x} = Ax + bu$$
$$y = Cx + Du$$

In standard control, we are trying to stabilize a system (or track a reference). This means finding critial points of a transfer function and finding the number of encirclements of the Nyquist plot. Its not enough to find a stable system. We also want to find the optimal trajectory for our variables.

Example 4.1. *Lets say we have a system that changes lanes for our car. There are several ways a car can change lanes.*

We have several parameters to track; fuel, time, comfort

The problem we are addressing with Optimal Control is:
textbfHow do we choose the best control input such that we have the best control system?

Essentially, this is what we mean by finding a performance measure. What variable in our inputs do we want to optimize? **MULTIVARIABLE CALCULUS**$\ddot{}$

Suppose we have the plant

$$\dot{x}(t) = f(x(t), u(t), t), x(t_0) = x_0$$
$$\text{And the cost function} \int_{t_0}^{t_1} L(x(t), u(t), t)dt + \phi(x(t_f))$$

Where

- L(*) measures the running cost of our system wrt to our input variables
- $\phi(*)$ measures the terminal cost, the error at the end to our reference value.

The optimal control problem is to find the input $u^*(t)$ ***on the time interval*** $[t_0, T]$ ***that drives the plant (3.2-1) along a trajectory*** $x^*(t)$ ***such that the cost function is minimized, and such that*** $\psi(x(T), T) = 0$ ***for a given function*** $\psi \in \mathbb{R}^p$

$x(t)$ is the trajectory, $x(T)$ is the terminal trajectory and ψ is the contraint (the $g(x)$ for the final trajectory)

4.1 Unconstrained optimization

$$min_{x \in \mathbb{R}^n} f(x)$$

where $f : \mathbb{R}^n \to \mathbb{R}$

x^* is the minimizer and $f(x^*)$ is a minimum we want to find. If we have a finite definition set, we are bound to find at least one local minima of f. If f is globally convex on $\mathbb{R}^n$, we have a global minimum point.

4.2 Constrained optimization

Want to optimize

$$min_{x\in\mathbb{R}^n} f(x) \text{ s.t. } g(x) = 0$$

where $f : \mathbb{R}^n \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}^p$

Think of lambda constraints. We have an equality constraint that arises such that

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T g(x)$$

For this equation to work, we have n lambdas. $\lambda \in \mathbb{R}^n$

To optimize we use the first order conditions (KKT)

- $\nabla\mathcal{L}(x^*, \lambda^*) = 0$
- $g(x^*) = 0$

Now we are bridging lagrangian equations to Optimal Control. We have a performance index.

$$J = \phi(x(T), T) + \int_{t_0}^{T} L(x(t), u(t), t)dt$$

subject to the constraints

$$\dot{x} = f(x(t), u(t), t) \text{ and } \phi(x(T), T) = 0$$

where $\psi \in \mathbb{R}^p$. $L(x, u)$ can be thougt of as the rate of cost for exerting control u in state x. And ϕ can be thougt of as the cost of ending up at state x

Our goal is to ***MINIMIZE*** the performance index J. This is done by finding the input $u^*(t)$ that drives the plant $x^*(t)$ along a trajectory that minimizes J such that the constraint $\psi(x(T), T) = 0$

4.2.1 Pontryagins maximum principle

Lev Pontryagin was developing this theory in 1956 working on rockets. It was derived using Bernoullis calculus of variations

Consider an n dim dynamic system with state variable $x \in \mathbb{R}^n$ and control variable $u \in \mathcal{U}$. The evolution of the dynamic system is a function of x and u (the state and control), according to the partial differential equation

$$\dot{x} = f(x, u)$$

The systems initial state is x_0 and has time interval $[0, T]$

The control trajectory is chosen to be $u : [0, T] \to \mathcal{U}$. The objective is a functional defined by

$$J = \psi(x(T), T) + \int_0^T L(x, u, t)dt$$

Where L is the rate of cost of exerting control u in state x.

The constrainst on the system dynamics come from the Lagrangian function $\mathcal{L}$ by introducing time varying constraint vector $\lambda(t) \in \mathbb{R}^n$, whose elements are called the costates of the system

This motivates the Hamiltonian. We now define the Hamtiltonian. It explores the KKT (first order conditions) for dynamic problems:

$$H(x, u, \lambda, t) = L(x, u, \lambda, t) + \lambda\,(t)^T\, f(x, u, t)$$

Pontryagins minimum principle states that the optimal state trajectory x^*, optimal control u^* and corresponding lagrange vector λ^* must minimize the Hamiltonian such that

$$H(x^*(t), u^*(t), \lambda^*(t), t) \leq H(x(t), u(t), \lambda(t), t)$$

- $\lambda(t)$ ***The costate***, and it enforces the dynamics constraint
- μ which is the terminal constraint. It enforces the terminal constraint

λ are the dual variables, if I nudge the state, λ tells me the impact

In practice, the Pontryagin maximum principle comes from a set of equations

- $\dot{x}^*(t) = \frac{\partial H}{\partial \lambda}(x^*, u^*, \lambda^*, t)$
- $\dot{\lambda}^*(t) = -\frac{\partial H}{\partial x}(x^*, u^*, \lambda^*, t$
- $\dot{u}^*(t) = argmin_{u \in \mathcal{U}} H(x^*, u^*, \lambda^*, t$

4.2.2 Augmented performance index

We start with our optimization problem

$$J = \phi(x(T), T) + \int_0^T L(x, u, t)dt$$

subject to the dynamics of

$$\dot{x} = f(x, u, t)$$

To enforce these constraints, we introduce time varying multipliers $\lambda(t)$, the costates and form the agumented performance index. We now define the augmented performance index

$$\hat{J} = \phi(x(T), T) + \mu^T \psi(x(T), T) + \int_{t_0}^T \left[L(x, u, t) + \lambda(t)^T \left(f(x, u, t) - \dot{x}(t) \right) \right] dt$$

This is the Lagrangian in Optimal Control, or in time varying dynamics

To understand the difference between static and dynamic optimization, look no further

NOTE: COME BACK TO THIS LATER AS THIS WAS CONFUSING

4.3 Bellmans principle of optimality

Definition 4.2. *Principle of optimality: An optimal policy has the property that, whatever the initial state and decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision*

Put simply. Consider a network of nodes. You start at node 0 and end up at the optimal node N. You have an optimaly policy that is the "best path from node 0 to node N. Then for every subnode within that optimal policy, is also an optimal node.

Example 4.3. *Travel from Oslo to Bergen. If Lillehammer is a part of the optimal path. Then from Lillehammer to bergen, you must find a new optimal path. So the new path becomes*

$$\textit{Oslo} \rightarrow \textit{Lillehammer} + \textit{Better Lillehammer} \rightarrow \textit{Bergen}$$

4.3.1 Idea of Dynamic programming

The algorithmic implementation of principle of optimality.

A global problem. Break the problem into smaller sub-problems. Optimize over all those subproblems and move recursively backwards to the initial state

4.4 Continuous control and the Hamilton-Jacobi-Bellman equation

Suppose we have a plant

$$\dot{x} = f(x, u, t)$$

And it has a performance index of

$$J(x(t_0), t_0) = \phi(x(T), T) + \int_{t_0}^{T} L(x, u, t)dt$$

Performance index is a cost functional telling you how well a control input performs, by combining accumulates running costs during the trajectory and the terminal cost at the final state.

We are interesting in finding the optimal control $u^*(t)$ on $[t_0, T]$ that optimizes J and drives the initial state $x(t_0)$ to the final state satisfying $\psi(x(T), T) = 0$

$$\psi : \mathbb{R}^n \rightarrow \mathbb{R}^p$$

Think of ψ as the constraint function $g(x)$ in the Lagrange equation

Now suppose we are travelling from state $x(t)$ to a small step ahead $x(t+\delta t)$. Then the cost to go $J(x(t),t)$ can be written as

$$J(x(t),t) = \phi(x(t),t) + \int_{t+\delta t}^{T} L(x,u,\tau)d\tau + \int_{t}^{t+\delta t} L(x,u,\tau)d\tau$$

We looked at KKT and Pontryagin maximum principle. We have under this chapter looked at the value function way, or the HJB. The Hamiltonian Jacobian Bellman function way. These two approaches are equivalent but different approaches to finding the optimal u^* such that the state $x(t) \to x^*(t)$ under the constraint $\psi(x(T),T)$

5 Optimal estimation

Reading: Crassidis & Junkins

- Kalman filters: 3.1-3.3
- Spacecraft dynamics: 6.1 appendix A.7-A.8
- Probability: Appendix C

5.1 Reachability